

Ashleigh Crosby

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Hacker, runner, tinkerer, wonderer.

I'm a problem-solver, an explorer, and a bit of an adventurer.

I think about systems on the grand and micro scale, I'm extremely quick to pick up on messy concepts and see the patterns as they emerge, and I'm always learning new skills.

I built a Wifi-enabled Ouija board as a prop for a horror LARP.

I can build furniture, bathrooms, bikes, and PCBs.

I repair my own clothes, which are frequently described as 'loud' and 'flamboyant as hell'.

What I'm looking for

Linux systems admin, cloud engineering, platform design and architecture.

What I've done

— **Vitrifi Ltd**

(2022-08-24 – 2025-12-11)

Platform Infrastructure Engineer

London (remote)

- Designed, deployed and maintained an on-premises cloud platform built on Apache CloudStack, using HAProxy, Puppet, and NodeJS to automate deployment on controllers, compute hosts, and network devices
- Automated DNS and BGP configuration within a CloudStack environment with the use of Python in conjunction with knot-resolver and exaBGP APIs
- Wrote numerous bugfixes and new features for the CloudStack Terraform provider
- Automated the installation and configuration of Prometheus exporters for a wide range of services and endpoints, with exporter service discovery for auto-configured monitoring
- Integrated new cloud platform with existing deployment mechanisms, using Ansible playbooks run by GitLab CI
- Automated the deployment of dozens of internal Linux-based systems, comprising an entire production delivery network
- Implemented firewalls, logging, monitoring, and SIEM integration with automated service discovery across the entire on-premises estate
- Migrated development and production systems from local users and groups to SSSD and PAM
- Investigated and demonstrated the integration of Cilium BGP with externally-managed WireGuard VPNs, allowing for natively-routed Kubernetes clusters spread across multiple PoPs

- Deployed Harbor with RadosGW backend for internal image storage and management
- Created a live embedded network monitoring map by polling gRPC endpoints and representing the results with JS
- Wrote management scripts for the automation of NVIDIA virtual devices, for MLOps purposes
- Deployed a fully-automated, tuned, highly-available, general-purpose Ceph storage cluster, with RadosGW for object storage, RDB for disk imaging, and CephFS for shared filesystems

— Unipart Digital

Senior Systems Engineer

(2018-10-01 – 2022-08-19)

Cambridge (hybrid)

- Wrote FreeIPA plugins to support tree changes that reflect internal structure for an organisation of 10,000+ users
- Managed three production OpenStack clusters, supporting ~1300 cores across three sites
- Extended OpenStack with Python and Bash to simplify networking and DNS at the user level
- Redesigned and deployed the monitoring and backup systems for hundreds of production systems
- Built Puppet modules for just about everything at UD
- Built plugins and config management for FreeIPA, Postfix, Dovecot to represent internal user structures

— PocketWorks

Lead Developer

(2018-01-15 – 2018-08-31)

Leeds

- Built a client-ready VR demo app in my first week, using Flutter and Google Cardboard
- Made significant UX and reliability improvements to a Rails-based UML diagram tool
- Aided in the design and conception of apps for new clients as part of a concerted marketing drive
- Implemented bugfixes and features to a long-running iOS / Rails sales app
- Written applications using standalone Eloquent, Sequelize, Hibernate, NHibernate and Doctrine ORMs against a variety of backends
- Built a prototype Android app to read and write complex data from STMicro's ST25DV NFC tags
- Developed a cross-platform VR demo using Flutter and Google Cardboard

— TempTribe

Lead Developer

(2017-06-01 – 2018-01-14)

London

- Dragged a legacy VB.NET staff management platform into the 21st century, with endless bugfixes, UI updates, features, security and performance improvements, and more

- Enabled the expansion of the company into international markets by porting and internationalising the entire backend, frontend and mobile platforms
- Delivered a brand new staff management mobile app for iOS and Android, built in React Native against a distributed Node.JS API and caching platform
- Sole day-to-day support of legacy and current systems for staff booking, job creation and payroll systems
- 4+ years' experience working with and developing .NET web and console applications in VB.NET and C#

— The Student Room Group

(2015-12-01 – 2017-05-31)

Senior Developer

Brighton

- Used Relay, GraphQL, and Memcached to build a scalable API for a forum with tens of millions of posts
- Delivered Apple and Android apps to their respective app stores, with experience in code-signing, user testing and delivery practices
- Built mobile-first web apps using React.JS, JSX and Apollo, embracing the latest language features in ES2015 and beyond
- Developed a Ruby / JS build system with full App Store integration, using nodegit and fastlane
- Experienced mentoring novice developers, having previously volunteered for CodeBar Brighton and helped budding coders into their first professional roles
- Project lead on TSR's mobile discussion app
- Integration of Firebase, Google Analytics, TestFairy, and Sentry APIs, to keep track of everything in the TSR codebase
- Written solid, test-driven code using Mocha, Chai and Enzyme, with Nuclide and React DevTools
- Built micro- and macro-services using Lumen and Laravel 5.* to support search and JSON API functionality for TSR's ever-growing services
- Augmented services with custom build scripts and continuous integration, built in Ruby and Java
- Wrote a complete BBCode parser in Node.JS, and accompanying renderer in React Native, to correctly render phpBB posts in a native mobile app
- Created extensive boilerplate code to integrate TSR with the Dynamic Yield personalisation platform
- Developed fast and scalable backend services in Node.JS
- Delivered scalable, testable Android code with extensive use of RxJava for observables and Dagger for dependency injection
- Tested JS and Node.JS code with Mocha / Chai, including promise support through Chai-as-promised
- Replaced the aging TSR search pages with a sleek and powerful system built on Google Search Appliances
- Conceived and built a rapid and clean discussion app for TSR, using React Native, Relay, GraphQL, and more
- Developed entire React Native component structure, navigation, styling, and higher-order components for a widely-used forum app

— **Omnisis Ltd**

(2012-07-01 – 2015-11-30)

Developer

Manchester

- Frontline support and sole full-time developer on a large .NET web application
- Supported a range of corporate clients through complex surveying needs, from local food companies to global pharmaceutical giants
- Implemented the AWS port of the entire surveying platform

— **Pingala Media Ltd**

(2011-07-01 – 2012-06-30)

Developer

Maidstone

- Managed and maintained all development and production servers
- Ported, maintained, and redeveloped a wide selection of PHP sites for schools and small enterprises
- Development of full inhouse CMS for new clients, based on the Symfony2 platform
- Migrated ~50 school and academy websites from a previous hosting provider to our own, with code fixes and improvements applied across the board
- Experienced with WordPress, Composer, Zend, Symfony, Doctrine, Eloquent, Magento - you name it, I've taken it apart

— **Other**

(1986-12-10 – 2026-02-05)

- Managed the logging of events from hundreds of production systems using a combination of rsyslog2, Elasticsearch and OpenDistro
- Implemented solutions against MongoDB for document-based storage, and Redis for key-value storage
- Extremely experienced working within Agile processes, adapting Scrum working practices to different cross-functional teams, tracking work in Jira / Confluence etc
- Experienced with Bootstrap for rapid application prototyping
- Developed against a range of caching solutions, including Varnish, Memcached, and node-cache
- Developed and hosted a range of personal sites, galleries, online game data crunching tools
- Built and automated icinga2 clusters to monitor all development and production systems, including custom checks for new services
- Worked extensively with MySQL and PostgreSQL for relational data persistence, including PL/SQL, sequencing and complex indexing
- Deployed replicating clusters for MySQL / MariaDB / PostgreSQL, using a variety of modern replication strategies, with Corosync / Pacemaker for automated cluster management
- Extremely confident at getting the best out of others, with a focus on efficient and accurate working, and using the right tools for the job
- Using Git for absolutely everything
- Built and maintained two iterations of a Drupal site for a 500-member uni rock society
- Written complex maintenance and automation scripts in Bash and Python
- Confident configuring, maintaining and integrating everything into CI platforms like ConcourseCI and CircleCI, as well as legacy systems like Jenkins

- Decades of experience with the packaging ecosystems for a wide range of modern distros (APT, DNF and the like) and languages (Composer, NPM, RubyGems, Bower and more)
- Created interactive charts, single-page applications, synthesized audio, procedurally-generated art, infinite-scroll galleries and more
- Worked extensively with modern JS APIs for responsiveness, async communication, JSON data encoding and more
- Built Docker images to run custom ConcourseCI resource types to streamline the automation of package builds
- Experienced with Apache and nginx for HTTP, including all kinds of authnz, and creating Puppet modules to automate everything
- Worked extensively with Puppet, PuppetDB, and Hiera, including writing my own ENC's, managing secrets at rest with hiera-eyaml, and creating modules for the management of Ceph, OpenStack, Apache Superset, Mediawiki, Taiga, Jitsi and more, with security concerns managed up front
- Built PCs, servers, home routers, office networks, programmable disco lights, even a Wifi-enabled ouija board
- A lifelong interest in security and fuzzing, with strong skills in intrusion detection, SSH configuration, key and certificate management, iptables and nftables, and SELinux
- ~18 years messing with networks, routers, switches, physical and virtual, at home and at work
- Developed a digital literacy tool for an education consultancy, handling massive datasets, animated graphing, file-based caching, map-reduce and PDF exports
- ~20 years' experience with Linux in a personal and professional context, building servers, configuring kernels, patching modules, automating configuration, and constantly learning

What I've (officially) learned

— **Keele University**

BA 2:1

English & Media

(2006 – 2011)